Problems using Pythagoras' rule

A Use Pythagoras' rule to calculate the length of the diagonal in each rectangle (to one decimal place).

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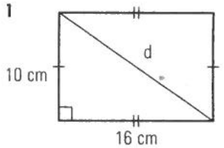

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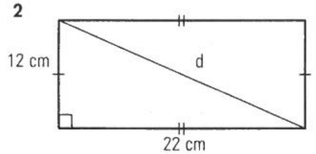

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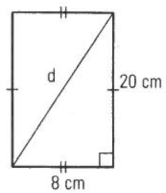

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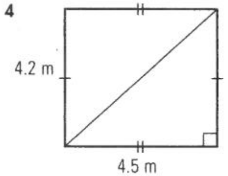

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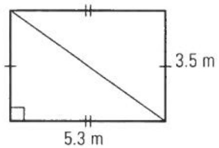

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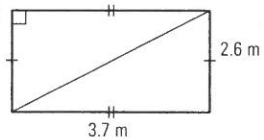
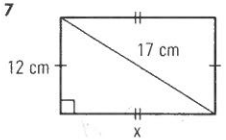


Calculate the length of each rectangle (correct to one decimal place).

7


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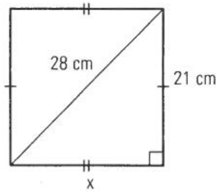

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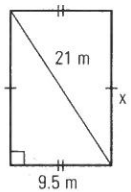
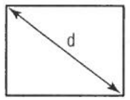


B

1 Television sizes are determined by measuring the diagonal of the screen. This screen is 42 cm wide and 32 cm high. Find the length of the diagonal, to the nearest centimetre.

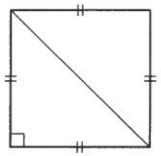


2 A television screen has a diagonal length of 65 cm. If the screen is 50 cm wide, what is its height? (Answer to the nearest centimetre.)

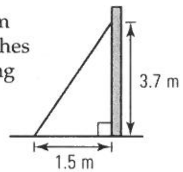
3 Find the length of a rectangle that has a diagonal measuring 210 mm and a width of 125 mm. Answer to the nearest millimetre.

4 Calculate the length of the diagonals of a square with sides measuring 35 cm. Answer correct to one decimal place.

5 A square has a diagonal of 65 cm. Calculate the length of its sides, to the nearest centimetre.



6 The foot of a ladder is 1.5 m from a vertical wall. It reaches 3.7 m up the wall. How long is the ladder? Answer to the nearest metre.



7 A ladder 4.5 m long is placed with its foot 1.8 m from the base of the wall. How far up the wall will the ladder reach? (Answer correct to one decimal place.)

8 A small ladder is leaning against a wall. The ladder reaches 138 cm up the wall. Its base is 45 cm from the wall. How long is the ladder? (Answer to the nearest centimetre.)

9 A gymnast performs a tumbling exercise along the diagonal of a 12-metre square floor. How far is it across the diagonal? (Answer correct to one decimal place.)

10 A square has a diagonal measuring one metre. Calculate the length of its sides, to the nearest centimetre.

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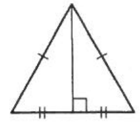
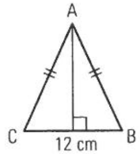
C

In an **isosceles** triangle, the altitude (vertical height) is an axis of symmetry. The altitude meets the base at right angles and cuts the base in half, forming two identical right-angled triangles.



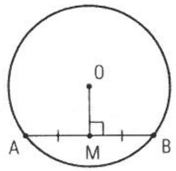
1 The base of an isosceles triangle is 12 cm long and the altitude is 15 cm. How long are the equal sides AB and AC? (Answer correct to one decimal place.)



2 An isosceles triangle has equal sides measuring 54 cm and a base of 36 cm. Calculate its vertical height (altitude) to the nearest centimetre.

3 The equal sides of an isosceles triangle measure 29 cm and its vertical height is 15 cm. Calculate the length of its base correct to one decimal place.

4 In a circle with centre O, M is the midpoint of a chord AB. The line OM is perpendicular to the chord. If the chord AB is 64 mm long and OM = 24 mm, calculate the radius of the circle, OA.

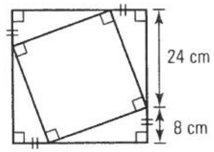


5 In a circle with radius 50 mm, the perpendicular distance from the centre of the circle to a chord measures 30 mm. How long is the chord?

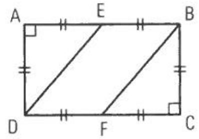
6 A chord measuring 42 mm is drawn in a circle with radius 29 mm. What is the distance from the chord to the centre of the circle?

7 An equilateral triangle has sides measuring 30 cm. Calculate its vertical height (altitude) correct to one decimal place.

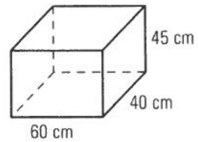
8 In this diagram, the larger square has sides measuring 32 cm. Calculate the perimeter of the smaller square, correct to one decimal place.



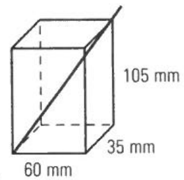
9 The rectangle ABCD is 12 cm long and 6 cm wide. Calculate the perimeter of DEBF, correct to one decimal place.



10 A box used to store firewood is the shape of a rectangular prism, 60 cm long, 40 cm wide and 45 cm deep inside. Calculate the length of the longest stick that can
a lie along the base of the box
b fit inside the box.
Give your answers to the nearest centimetre.



11 A fruit juice container has the shape of a rectangular prism. It needs a straw that must extend 25 mm beyond the container while touching the furthest corner of the base. Calculate the minimum length of the straw needed, correct to the nearest millimetre.



12 A soft drink comes in a cylindrical can, 65 mm in diameter and 125 mm high. A straw 205 mm long is placed in the can with one end touching the base. What is the minimum length that the straw can extend ouside the can? Give your answer to the nearest millimetre.

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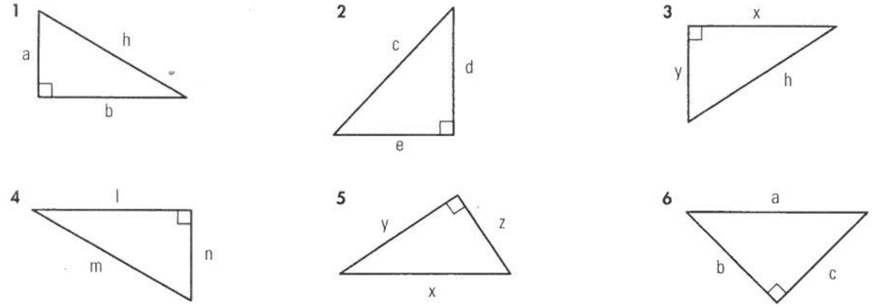
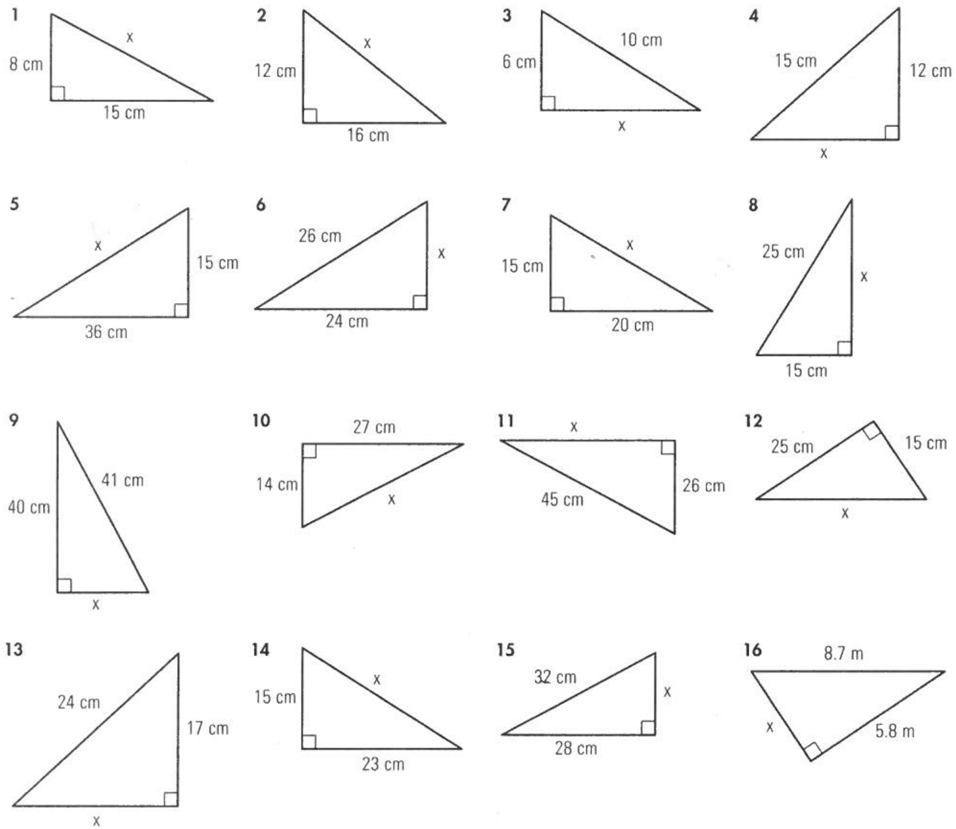
Pythagoras' rule: mixed calculations

A Use the given pronumerals to write Pythagoras' rule for each triangle.



B Calculate the length of side **x** in each triangle (correct to one decimal place).



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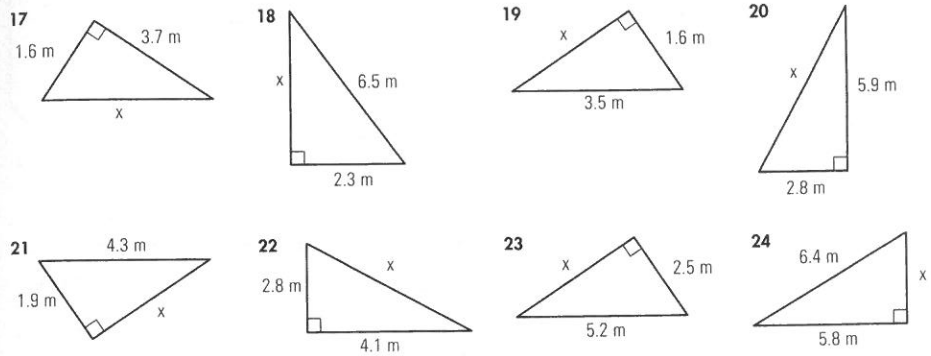
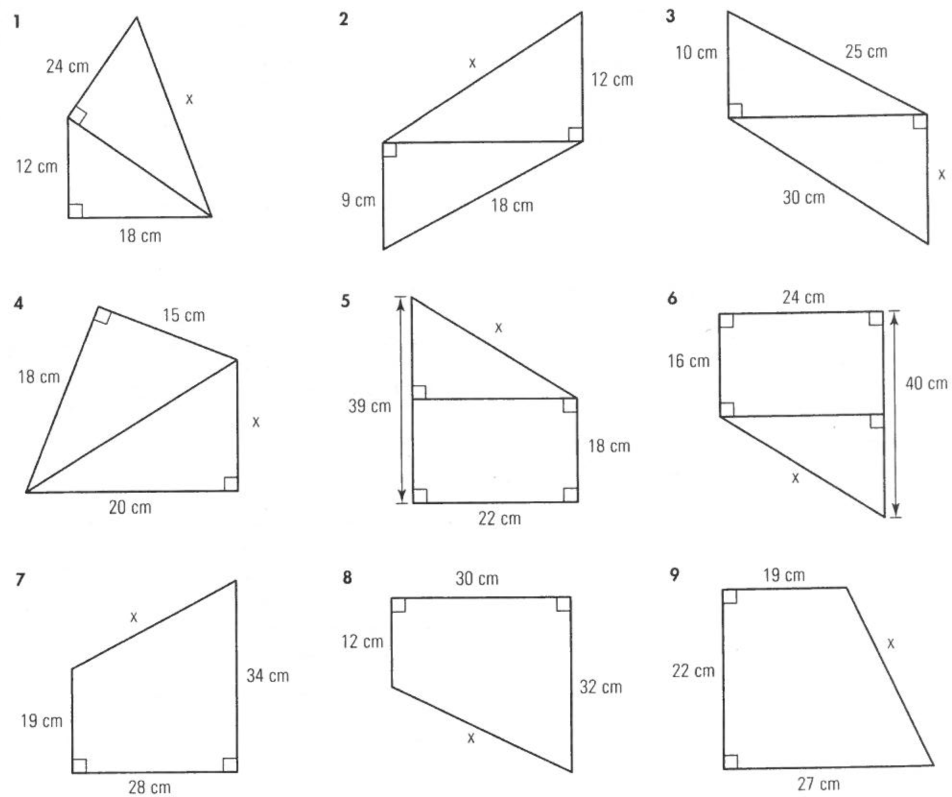
Calculate the length of side **x** in each triangle, correct to one decimal place.



C Find the value of **x** in each diagram, correct to one decimal place.



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